Practical no. 19

Aim - Cross-Browser Testing Using Selenium Grid

Execute automation scripts across multiple browsers (Chrome, Firefox, Edge) using Selenium Grid. Compare execution time and compatibility issues.

Step 1: Check Java

Open Command Prompt and run:

```
java -version
mvn –version
```

Step 2: Create a Selenium Grid Folder

In Command Prompt run:

```
mkdir C:\SeleniumGrid
cd /d C:\SeleniumGrid
```

Step 3: Download Selenium Server

Download the Selenium Server 4.35.0 JAR (selenium-server-4.35.0.jar) and put it inside C:\SeleniumGrid

Check it by running:

```
dir C:\SeleniumGrid
```

Step 4: Start Selenium Grid

In the same Command Prompt:

```
cd /d C:\SeleniumGrid
java -jar selenium-server-4.35.0.jar standalone
```

Do not close this window. Selenium Grid starts on http://localhost:4444

Step 5: Check Selenium Grid in Chrome

Open Chrome and enter in the address bar: http://localhost:4444

You should see the Selenium Grid page.

Step 6: Open Your Eclipse Project

File → New → Maven Project

Tick: Create a simple project → Next

Step 7: Enter Project Details

Group Id: com.example

Artifact Id: selenium-project

Leave the other settings as default → Finish

Step 8: Open pom.xml

In Project Explorer, double-click pom.xml and replace its contents with (make sure Selenium is included):

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
   https://maven.apache.org/xsd/maven-4.0.0.xsd">
   <modelVersion>4.0.0</modelVersion>
   <groupId>com.selenium</groupId>
   <artifactId>selenium-project</artifactId>
   <version>1.0-SNAPSHOT</version>
   <properties>
```

```
      <maven.compiler.source>21</maven.compiler.source>
      <maven.compiler.target>21</maven.compiler.target>
   </properties>
   <dependencies>
      <dependency>
         <groupId>org.seleniumhq.selenium</groupId>
         <artifactId>selenium-java</artifactId>
         <version>4.35.0</version>
      </dependency>
      <dependency>
         <groupId>org.testng</groupId>
         <artifactId>testng</artifactId>
         <version>7.11.0</version>
         <scope>test</scope>
      </dependency>
   </dependencies>
</project>
```

Save – Ctrl + S

Step 9: Create Package

Right-click src/test/java → New → Package

Enter: tests → Finish

Step 10: Create Java Class

Right-click tests → New → Class

Enter: LoginTest → Finish

Open LoginTest.java and replace its contents with the following (Chrome through Grid):

```
package tests;
import java.net.URL;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.remote.RemoteWebDriver;
import org.testng.Assert;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;
public class LoginTest {
   WebDriver driver;
   @BeforeMethod
   public void setUp() throws Exception {
      ChromeOptions options = new ChromeOptions();
      driver = new RemoteWebDriver(
         new URL("http://localhost:4444"),
         options
      );
   }
   @Test
   public void verifyGoogleTitle() {
      long startTime = System.currentTimeMillis();
      driver.get("https://www.google.com");
      String title = driver.getTitle();
      long endTime = System.currentTimeMillis();
      System.out.println("================================");
      System.out.println("Browser: Chrome");
      System.out.println("Page Title: " + title);
      System.out.println("Execution Time: "
         + (endTime - startTime) + " ms");
      System.out.println("================================");
      Assert.assertEquals(title, "Google");
   }
   @AfterMethod
   public void tearDown() {
      if (driver != null) {
         driver.quit();
      }
   }
```

```
}
```

Step 11: Run the Chrome Test

Right-click LoginTest.java → Run As → TestNG Test

Step 12: Firefox

Open LoginTest.java and replace the code with:

```
package tests;
import java.net.URL;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.remote.RemoteWebDriver;
import org.testng.Assert;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;
public class LoginTest {
   WebDriver driver;
   @BeforeMethod
   public void setUp() throws Exception {
      FirefoxOptions options = new FirefoxOptions();
      driver = new RemoteWebDriver(
         new URL("http://localhost:4444"),
         options
      );
   }
   @Test
   public void verifyGoogleTitle() {
      long startTime = System.currentTimeMillis();
      driver.get("https://www.google.com");
      String title = driver.getTitle();
      long endTime = System.currentTimeMillis();
      System.out.println("================================");
      System.out.println("Browser: Firefox");
      System.out.println("Page Title: " + title);
      System.out.println("Execution Time: "
         + (endTime - startTime) + " ms");
      System.out.println("================================");
      Assert.assertEquals(title, "Google");
   }
   @AfterMethod
   public void tearDown() {
      if (driver != null) {
         driver.quit();
      }
   }
}
```

Step 13: Run Firefox

Right-click LoginTest.java → Run As → TestNG Test

Step 14: Edge

Replace the existing code in the same Java file with:

```
package tests;
import java.net.URL;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.edge.EdgeOptions;
import org.openqa.selenium.remote.RemoteWebDriver;
import org.testng.Assert;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;
public class LoginTest {
   WebDriver driver;
   @BeforeMethod
   public void setUp() throws Exception {
      EdgeOptions options = new EdgeOptions();
```

```
      driver = new RemoteWebDriver(
         new URL("http://localhost:4444"),
         options
      );
   }
   @Test
   public void verifyGoogleTitle() {
      long startTime = System.currentTimeMillis();
      driver.get("https://www.google.com");
      String title = driver.getTitle();
      long endTime = System.currentTimeMillis();
      System.out.println("================================");
      EdgeOptions options = new EdgeOptions();
      System.out.println("Page Title: " + title);
      System.out.println("Execution Time: "
         + (endTime - startTime) + " ms");
      System.out.println("================================");
      Assert.assertEquals(title, "Google");
   }
   @AfterMethod
   public void tearDown() {
      if (driver != null) {
         driver.quit();
      }
   }
}
```

Step 15: Run Edge

Right-click LoginTest.java → Run As → TestNG Test

Step 16: Add All Three Browsers into ONE Test

Replace LoginTest.java with:

```
package tests;
import java.net.URL;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.edge.EdgeOptions;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.remote.RemoteWebDriver;
import org.testng.Assert;
import org.testng.annotations.Test;
public class LoginTest {
   @Test
   public void chromeTest() throws Exception {
      ChromeOptions options = new ChromeOptions();
      WebDriver driver = new RemoteWebDriver(
         new URL("http://localhost:4444"),
         options
      );
      long start = System.currentTimeMillis();
      driver.get("https://www.google.com");
      String title = driver.getTitle();
      long end = System.currentTimeMillis();
      System.out.println("Chrome");
      System.out.println("Title: " + title);
      System.out.println("Time: " + (end - start) + " ms");
      Assert.assertEquals(title, "Google");
      driver.quit();
   }
   @Test
   public void firefoxTest() throws Exception {
      FirefoxOptions options = new FirefoxOptions();
      WebDriver driver = new RemoteWebDriver(
         new URL("http://localhost:4444"),
         options
      );
      long start = System.currentTimeMillis();
      driver.get("https://www.google.com");
      String title = driver.getTitle();
```

```
      long end = System.currentTimeMillis();
      System.out.println("Firefox");
      System.out.println("Title: " + title);
      System.out.println("Time: " + (end - start) + " ms");
      Assert.assertEquals(title, "Google");
      driver.quit();
   }
   @Test
   public void edgeTest() throws Exception {
      EdgeOptions options = new EdgeOptions();
      WebDriver driver = new RemoteWebDriver(
         new URL("http://localhost:4444"),
         options
      );
      long start = System.currentTimeMillis();
      driver.get("https://www.google.com");
      String title = driver.getTitle();
      long end = System.currentTimeMillis();
      System.out.println("Edge");
      System.out.println("Title: " + title);
      System.out.println("Time: " + (end - start) + " ms");
      Assert.assertEquals(title, "Google");
      driver.quit();
   }
}
```

Save the code and run: Run As → TestNG Test

Step 17: If localhost:4444 Cannot Be Reached

Go back to your Grid Command Prompt and run:

```
cd /d C:\SeleniumGrid
java -jar selenium-server-4.35.0.jar standalone
```

Keep it open.